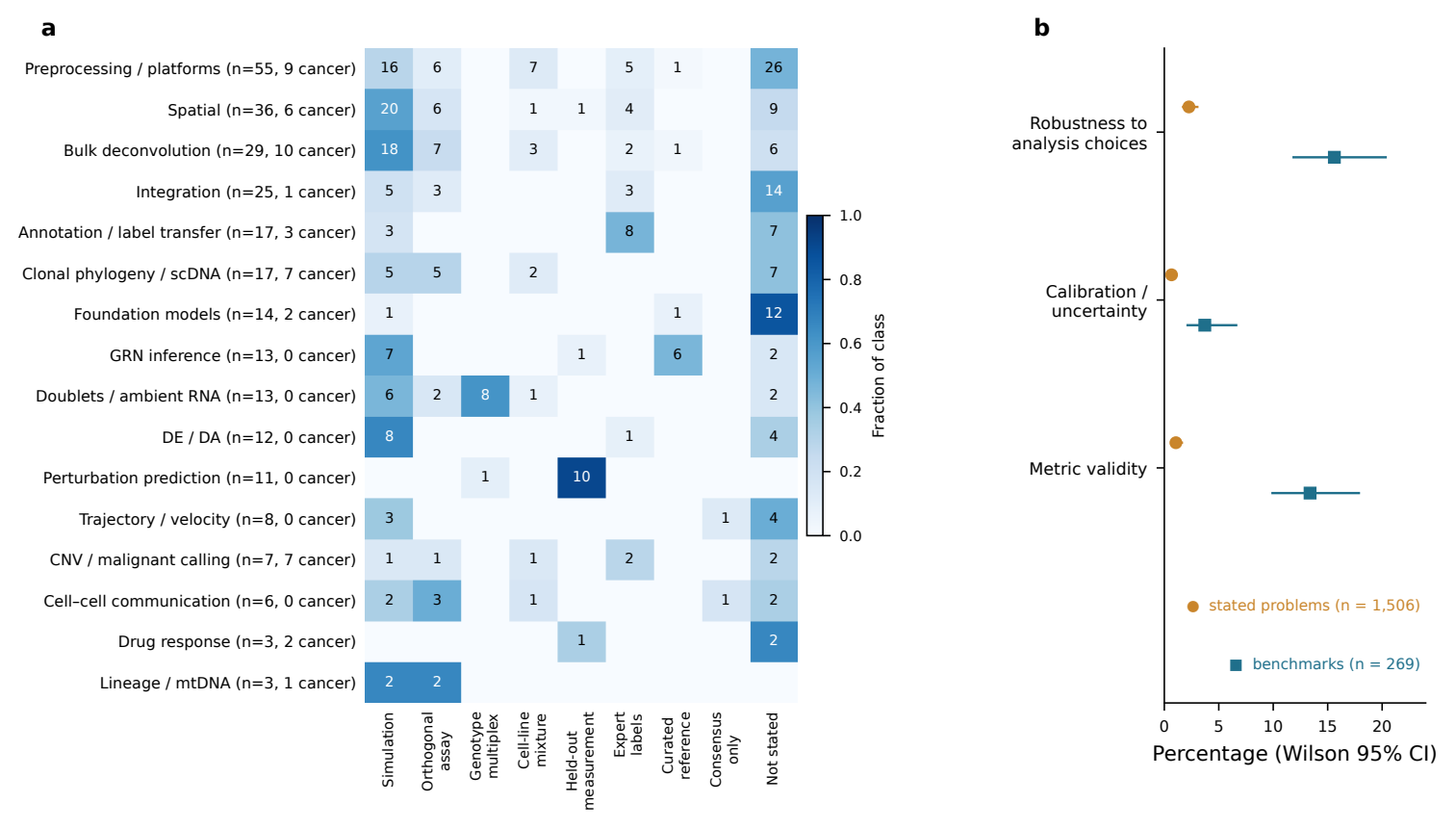
a
Preprocessing / platforms (n=55, 9 cancer)
Spatial (n=36, 6 cancer)
Bulk deconvolution (n=29, 10 cancer)
Integration (n=25, 1 cancer)
Annotation / label transfer (n=17, 3 cancer)
Clonal phylogeny / scDNA (n=17, 7 cancer)
Foundation models (n=14, 2 cancer)
GRN inference (n=13, 0 cancer)
Doublets / ambient RNA (n=13, 0 cancer)
DE / DA (n=12, 0 cancer)
Perturbation prediction (n=11, 0 cancer)
Trajectory / velocity (n=8, 0 cancer)
CNV / malignant calling (n=7, 7 cancer)
Cell–cell communication (n=6, 0 cancer)
Drug response (n=3, 2 cancer)
Lineage / mtDNA (n=3, 1 cancer)
Simulation
Orthogonal assay
Genotype multiplex
Cell-line mixture
Held-out measurement
Expert labels
Curated reference
Consensus only
Not stated
Fraction of class
1.0
0.8
0.6
0.4
0.2
0.0
b
Robustness to analysis choices
Calibration / uncertainty
Metric validity
stated problems (n = 1,506)
benchmarks (n = 269)
0
5
10
15
20
Percentage (Wilson 95% CI)